MACHINE LEARNING PROJECT

PROBLEM STATEMENT:

To examine whether increases in global crude oil prices, coupled with depreciation of the Indian Rupee (USD-INR), significantly transmit into Wholesale Price Inflation (WPI – Fuel & Power) and subsequently Consumer Price Inflation (CPI) in India.

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Transmission of Global Oil & Currency Shocks to Indian Inflation (2020-2024)

Quantifying inflationary pass-through, lag structures, and policy response windows.



Currency depreciation is the dominant driver of CPI inflation, necessitating a shift in policy priority toward forex stabilization.

MAGNITUDE GAP

2.2x

Impact of 10% INR shock vs. 20% Oil shock.

POLICY WINDOW



Lag between wholesale shock and peak consumer impact.

CORRELATION STRENGTH



COMBINED RISK

+9.9%

WPI Impact of Simultaneous Shock.

Executive Summary: While crude oil dominates wholesale costs, the exchange rate is the stronger lever for consumer inflation. Policy must pivot from broad rate hikes to targeted forex defense.

Source: MB25056_A1_Report, Executive Summary.

India faces a structural ‘Double Whammy’: 85% import dependency exacerbates vulnerability to global volatility



The convergence of energy prices and currency weakness creates sustained inflationary pressure, threatening the RBI’s 4% (+/- 2%) target.

Source: Ministry of Petroleum & Natural Gas; RBI Handbook of Statistics (2024).

A rigorous 7-stage analytical framework ensures high predictive accuracy (93% for CPI).





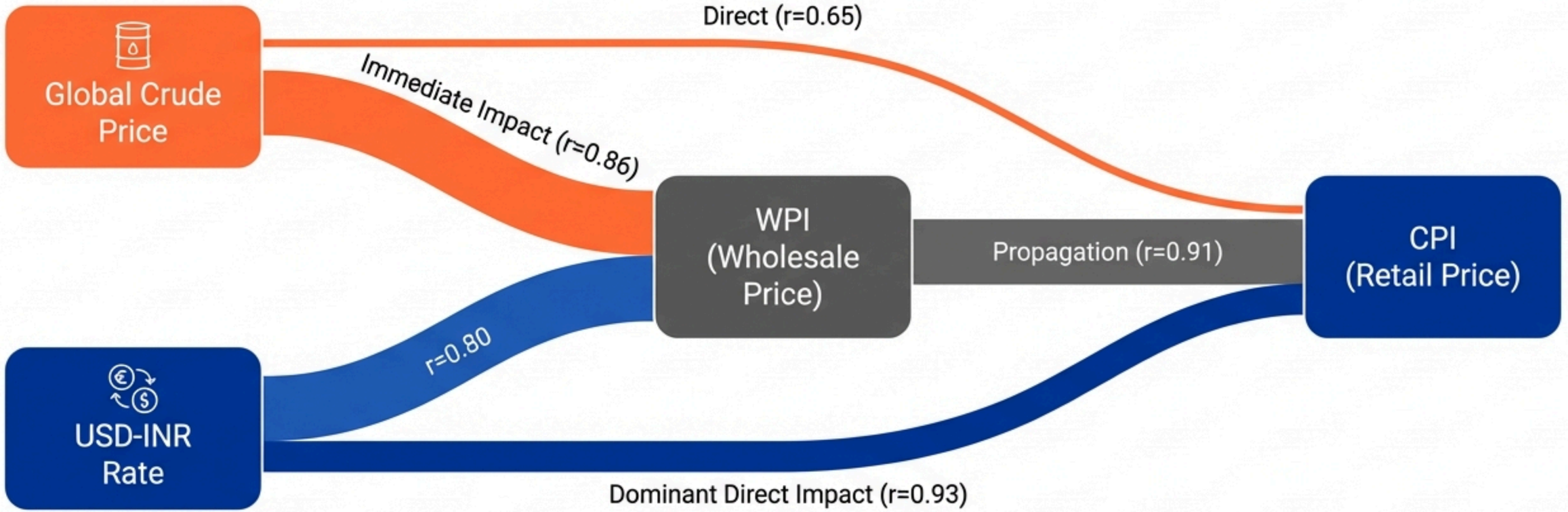
- **Data:** 48-month dataset (April 2020–March 2024) harmonized across Refinery, Global Price, and Inflation metrics.
- **Technique:** OLS Regression and Time-Lagged Correlation Analysis.
- **Validation**: Results cross-checked against IMF and World Bank secondary data.

Key Takeaway Box

Moving beyond simple observation to predictive modeling allows for high-confidence policy recommendations.

Source: Report Section 'Data and Methodology', Stages 1-7.

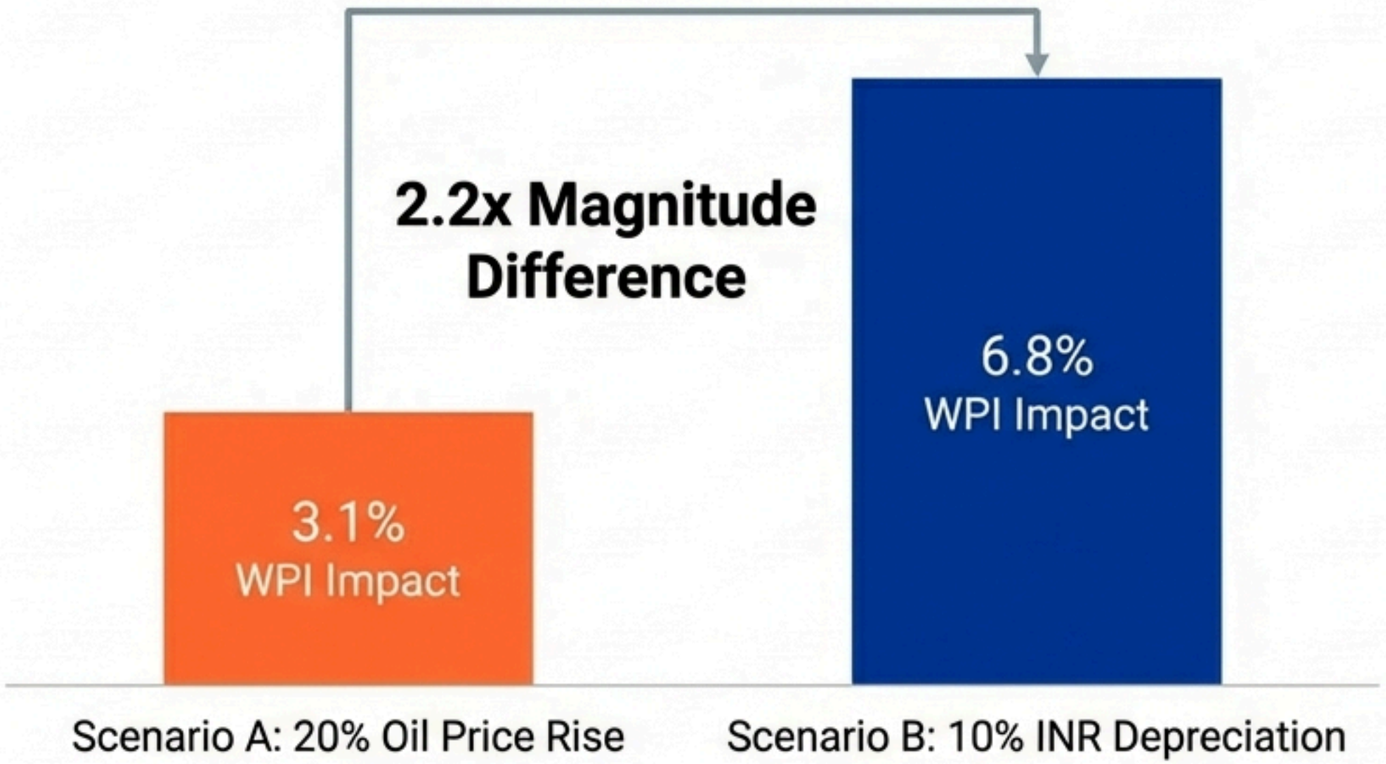
Inflation transmits hierarchically: External shocks hit WPI immediately, then propagate to CPI.



WPI acts as a leading indicator, but Currency (INR) has a massive direct bypass line to the Consumer (CPI) that is stronger than the oil price itself.

Source: Table 3: Transmission Path Correlation Analysis.

Counter-Intuitive Insight: Currency depreciation is 2.2x more inflationary than oil price shocks.

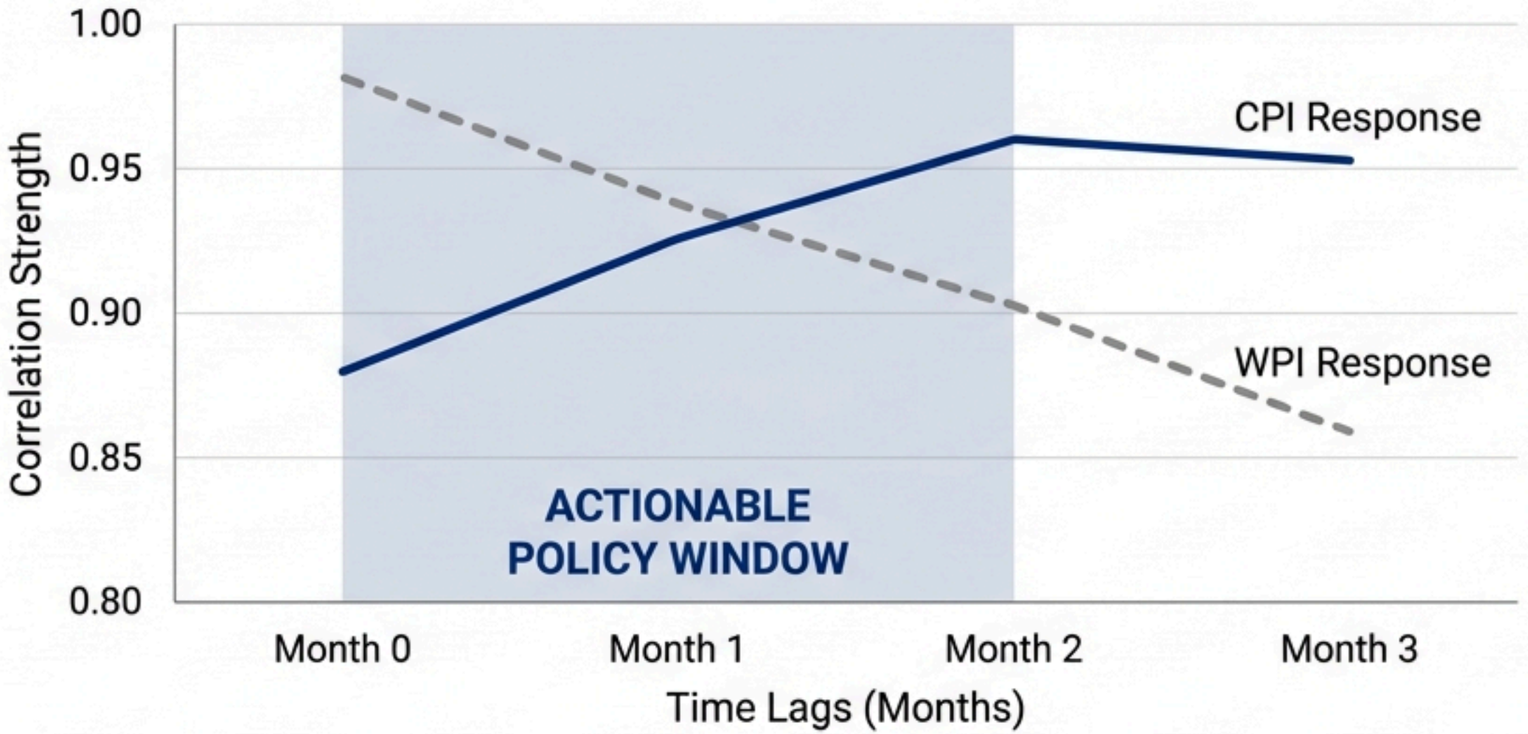


Why? INR depreciation affects the *entire* import basket (electronics, edible oils, machinery), not just energy inputs.

Key Takeaway Box: A 10% currency slip hurts the economy significantly more than a 20% oil spike. Controlling the exchange rate is the most effective inflation lever.

Source: Scenario Analysis Results (Scenario A vs Scenario B).

A 1-2 month ‘Policy Window’ exists between the external shock and peak consumer inflation



⚠ **Immediate WPI:** Wholesale prices adjust instantly to spot markets.

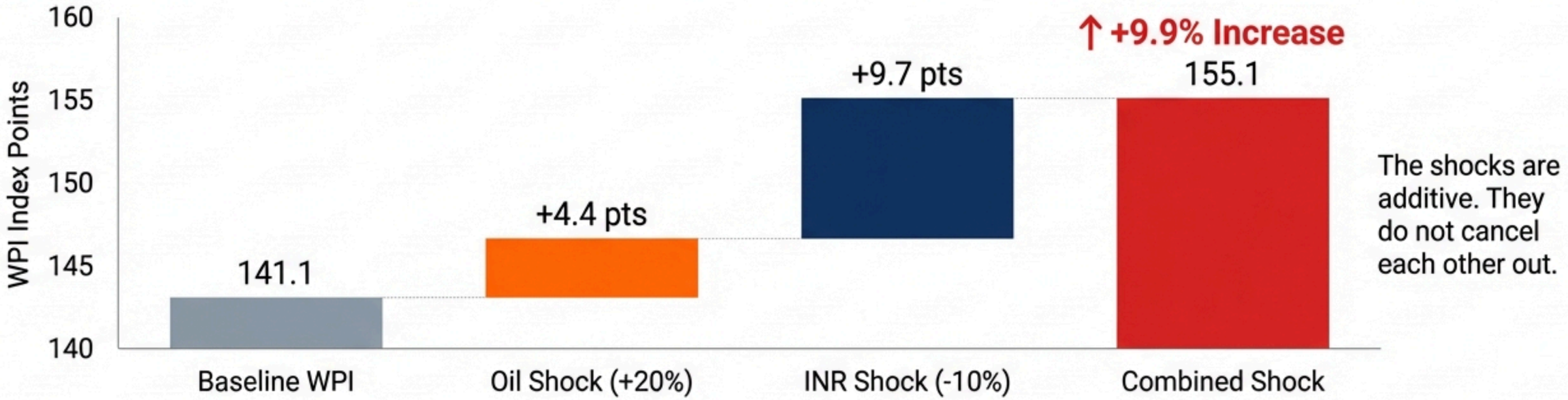
🕒 **Delayed CPI:** Retail prices take ~60 days to fully reflect the shock.

↗ **The Opportunity:** Authorities have 60 days to intervene before the household budget is hit.

Timing is everything. Reactive policy after Month 2 is too late.
The RBI must strike while the shock is still in the wholesale phase.

Source: Table 4 & 5: Lag Correlation Structures.

A ‘Perfect Storm’ combined shock could drive Wholesale Inflation (WPI) up by nearly 10%.



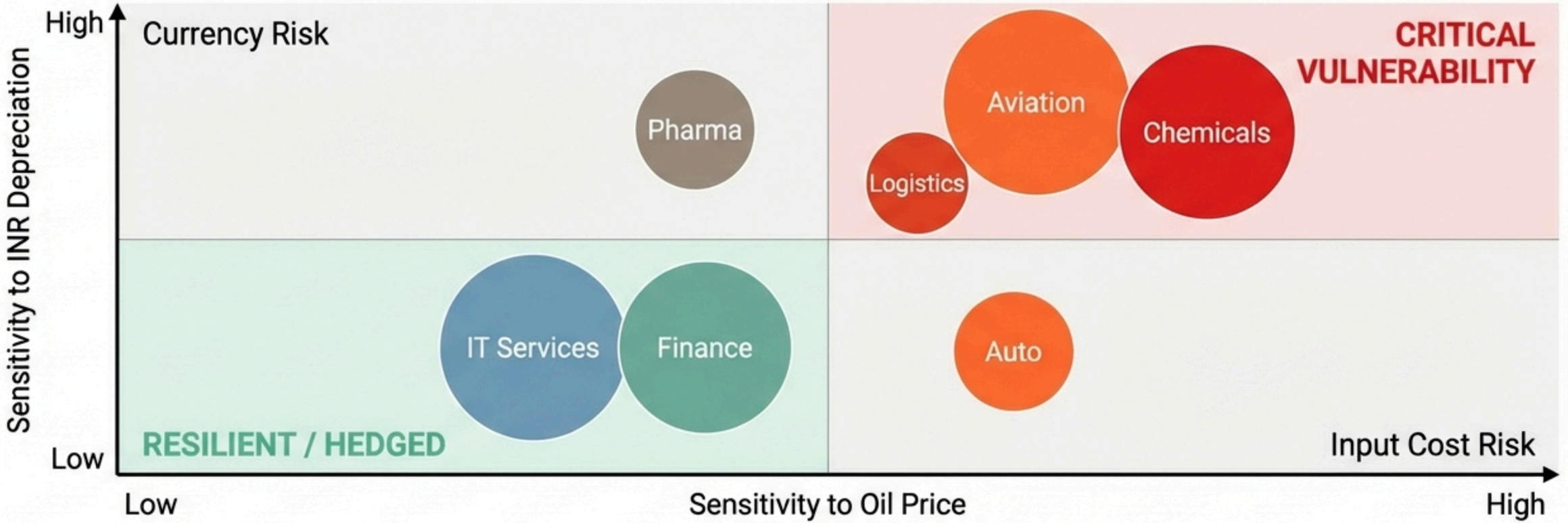
The shocks are additive. They do not cancel each other out.

- Scenario A **(Oil Only):** +3.1% Impact.
- Scenario B (INR Only): +6.8% Impact.
- Scenario C (Combined): +9.9% Critical Risk.

Simultaneous supply cuts and capital flight create a non-linear threat that breaks standard inflation targeting frameworks.

Source: Empirical Results - Scenario Analysis (A, B, C).

Risk exposure varies: Aviation and Chemicals face critical dual-threats, while IT remains resilient.



Strategy must be sector-specific. Aviation is in the 'kill zone', while IT Services acts as a natural portfolio hedge due to export earnings.

Source: Table 8: Sectoral Vulnerability Assessment.

A coordinated 3-Pillar Framework is required to break the inflation transmission spiral.

RBI (Monetary)



Forex Defense > Rate Hikes

Prioritize INR stability to cut the strongest transmission link (r=0.93).

Ministry of Finance (Fiscal)



Dynamic Excise & SPR

Trigger automatic excise cuts if Oil >$85/bbl. Release Strategic Petroleum Reserves.

Petroleum Ministry (Structural)



DBT Transition

Move from product subsidies to Direct Benefit Transfer to reduce leakage.

Bifurcated Approach: The RBI defends the currency (CPI), while the Finance Ministry buffers the wholesale cost (WPI).

Source: Discussion and Policy Implications; Section 2 & 3.

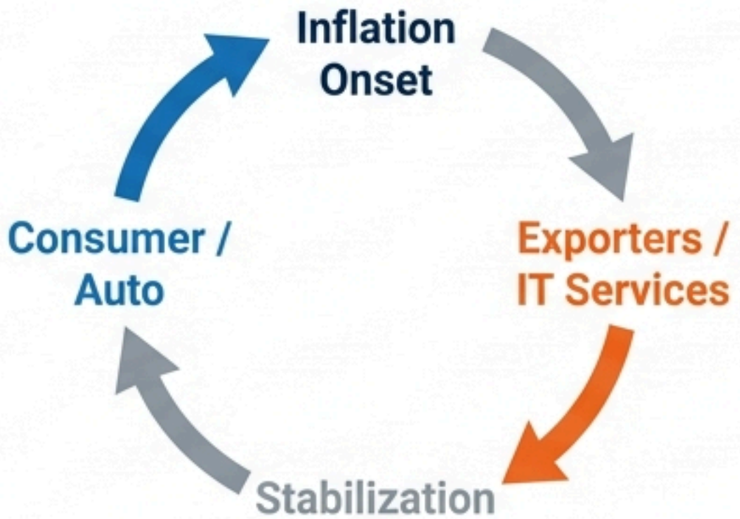
Businesses must adopt a 3-month rolling hedge horizon to align with the transmission cycle

The Hedging Calendar



Rolling 3-month forwards cover the full transmission window.

Sector Rotation Strategy



Rotate exposure based on the 1-2 month lag signal.

Don’t over-hedge. The data shows transmission completes in one quarter. A 3-month rolling window is the mathematical optimum.

Source: Corporate Strategy Implications

Strategic Imperative: Prioritize currency stability to safeguard the inflation target.

Validation ($p < 0.001$)



INR Impact > Oil Impact

60-Day Policy Window

Sector Asymmetry

CALL TO ACTION

IMMEDIATE
Activate **Forex Defense** if INR slips >5%.

TACTICAL
Release SPR if Oil >$85/bbl.

STRATEGIC
Accelerate Renewable Transition to reduce 85% dependency.

Final Verdict: We cannot control global oil prices, but we can manage our currency. That is where the battle for inflation will be won.

Source: Conclusions & Future Research Directions.